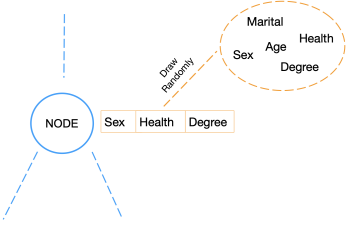
Introduction to Machine Learning

Random Forest In a Nutshell

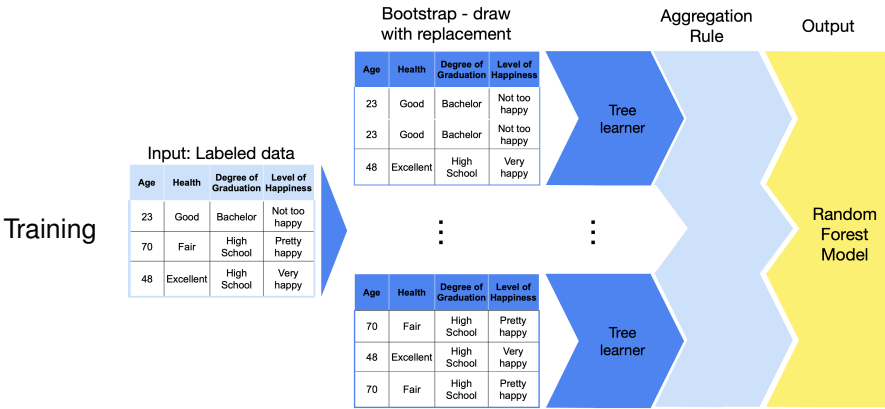


Learning goals

- Understand basic concept of random forest
- Know basic aggregation rules
- Understand concept of feature importance

LEARNING AND PREDICTION WITH RF

- Stabilizes tree learner by bagging (bootstrap aggregation)
- Randomizes tree learner and combines models into one meta model
- Can be adapted to learning task, i.e., classification or regression



LEARNING AND PREDICTION WITH RF

Prediction

Input: Unlabeled data

Age	Health	Degree of Graduation	Level of Happiness
41	Fair	Bachelor	?
35	Good	Bachelor	?
22	Fair	High School	?

Random Forest Model

Prediction

Level of Happiness
Not too happy
Pretty happy
Not too happy

AGGREGATION RULES FOR DIFFERENT TASKS

Classification Task - Majority Vote

Age	Health	Degree of Graduation	Level of Happiness
41	Fair	Bachelor	?

Tree | Tree | Tree | Tree | Tree

Pretty Happy | Not Too Happy | Not Too Happy | Not Too Happy | Pretty Happy

RF Prediction

Not Too Happy

Regression Task - Averaging

Rating	Income	Credit Limit	Credit Card Balance
107	32.318	4351	?

Tree | Tree | Tree | Tree | Tree

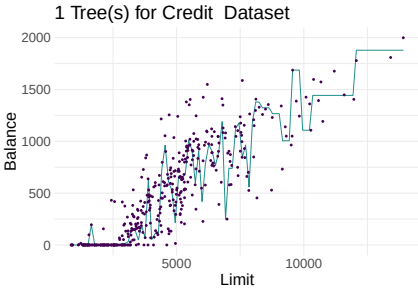
482 | 471 | 501 | 478 | 489

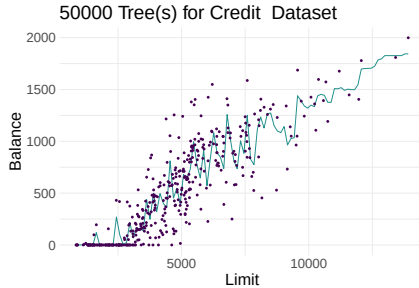
RF Prediction

484,2

PERFORMANCE OF RF

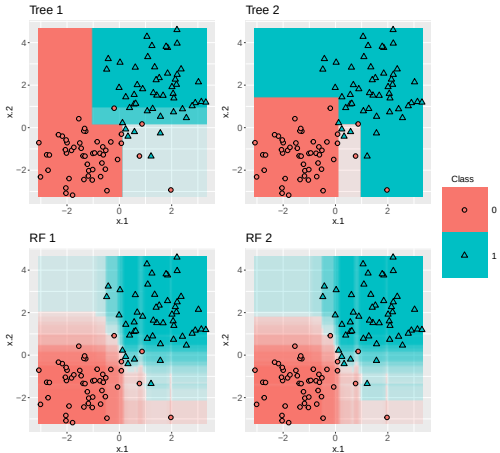
- In general: Increasing the ensemble size stabilizes the predictions
  - For regression tasks the stabilization is often not sufficient.





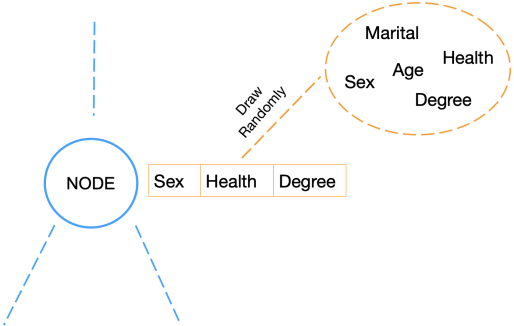
PERFORMANCE OF RF

- RF performs well for classification tasks:
  - Two different trees →Quite different decision regions
  - Two different RFs →Similar decision regions



PERFORMANCE OF RF

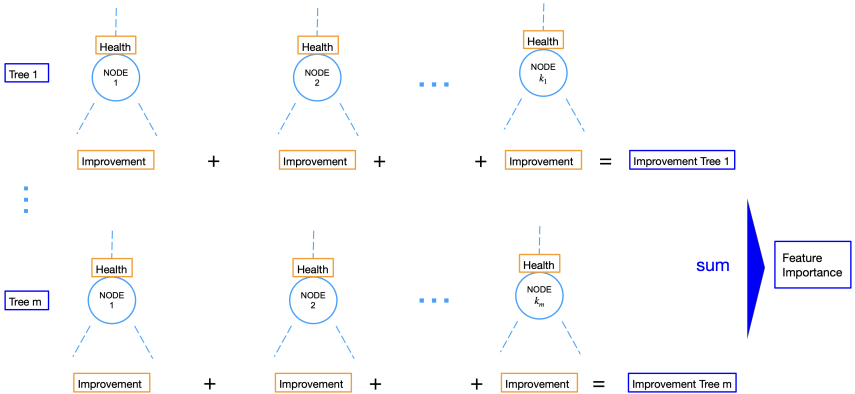
- Trees should be decorrelated, i.e., make mistakes in different directions
- Avoid correlation by
  - Bootstrap sampling
  - Randomized splits. In each node of each tree, consider different features for splitting:



FEATURE IMPORTANCE

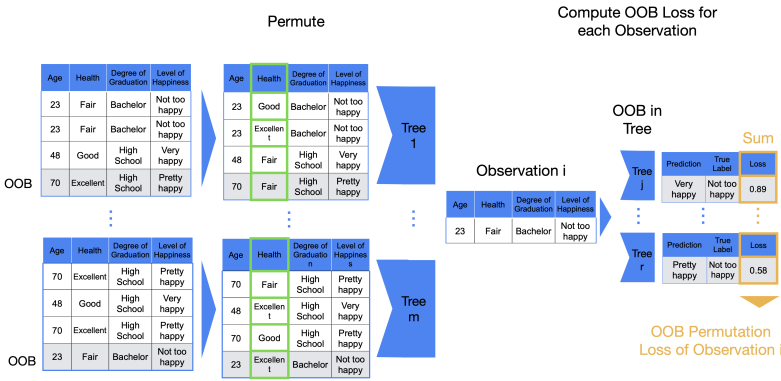
Several options, e.g., measure contribution of feature to model:

- Measure based on improvement in splitting criterion
- E.g. Feature importance of 'Health', search all nodes with 'Health' as splitting variable:

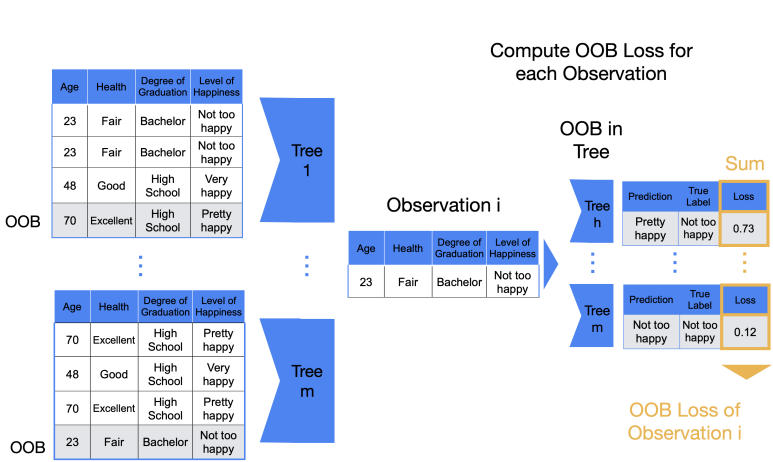


FEATURE IMPORTANCE

- Measure based on OOB Loss



FEATURE IMPORTANCE



Feature Importance 'Health' = **OOB Permutation Loss of all observations** - **OOB Loss of all observations**